

ECE 3337: Test #2

Fall 2025

Last name of student Solutions	Student ID number
First name of student	Email address

Do This First:

1. Make sure that you have all the pages of the exam.
2. Fill in the above information.
3. Turn off and put away all electronic gadgets (laptop, tablet, smartphone, calculator, smart watch, ...)

Exam Rules:

1. You are allowed to bring one 8.5" x 11" handwritten crib sheet (double sided).
2. Indicate your answers in the spaces provided. Use the back of the page to carry out calculations.

Q1	Q2	Q3	Q4	Q5	Q6	Total
20	20	20	20	10	10	100
Laplace Transform	Inverse Laplace Transform	Circuit Analysis	Response Partitions	Op Amp Circuits	Block Diagrams	

1. (20 points) Laplace Transforms

13+ for each

a. (6 points) Find $X(s)$ the Laplace transform of the following signal:

$$x(t) = 2\delta(t-3) + (2t-1)^3 e^{-2t} u(t-3) + t \sin(2t) u(t).$$

6 total

$$\textcircled{1} \mathcal{L}\{2\delta(t-3)\} = 2 \cdot e^{-3s}$$

1 pt

$$\textcircled{2} \mathcal{L}\{(2t-1)^3 e^{-2t} u(t-3)\} = e^{-3s} \mathcal{L}\{(2t+5)^3 - 1\} \cdot e^{-3s} = \frac{e^{-3s}}{e^s} \mathcal{L}\{(2t+5)^3 - 1\} \cdot e^{-2s}$$

$$(2t+5)^3 = 8t^3 + 60t^2 + 150t + 125$$

$$\textcircled{2} = \frac{e^{-3s}}{e^s} \left[\frac{8 \cdot 3!}{(s+2)^4} + \frac{60 \cdot 2!}{(s+2)^3} + \frac{150 \cdot 1!}{(s+2)^2} + \frac{125}{s+2} \right]$$

1 pt

1 pt

$$\textcircled{3} \mathcal{L}\{t \sin 2t\} =$$

$$\text{we } \mathcal{L}\{t \cdot f(t)\} = -\frac{d}{ds}[F(s)] \quad F(s) = \mathcal{L}\{\sin 2t\} = \frac{2}{s^2+4}$$

$$\textcircled{3} = \frac{4s}{(s^2+4)^2}$$

1 pt

$$2e^{-3s} + \frac{e^{-3s}}{e^s} \left[\frac{8 \cdot 3!}{(s+2)^4} + \frac{60 \cdot 2!}{(s+2)^3} + \frac{150 \cdot 1!}{(s+2)^2} + \frac{125}{s+2} \right] + \frac{4s}{(s^2+4)^2}$$

(b) (6 points) Given a signal $x_2(t)$, find the Laplace transform of $y(t) = 7e^{-t} \int_0^t x_2(\tau/2) d\tau$ in terms of $X_2(s)$

$$\text{Scaling } \mathcal{L}\{x_2(t/2)\} = \frac{1}{2} X_2\left(\frac{s}{2}\right)$$

$$\text{integration } \mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$$

$$\text{exponential } \mathcal{L}\{e^{-at} f(t)\} = F(s+a)$$

$$Z(t) = \int_0^t x_2(\tau/2) d\tau \quad \mathcal{L}\{Z(t)\} = \frac{1}{s} \mathcal{L}\{x_2(t/2)\} = \frac{2}{s} X_2\left(\frac{s}{2}\right)$$

$$\mathcal{L}\{e^{-t} Z(t)\} = Z(s+1) \quad Y(s) = \frac{7 \cdot 2}{s+1} X_2\left(\frac{s+1}{2}\right)$$

1 pt for each

1 pt

13+ for trying

13 + for trying

c. (8 points) Find the Laplace transform of $z(t) = t e^{-t} \cos(2t)u(t)$, and find the values of all its poles and zeros.

$\mathcal{L}\{t \cdot f(t)\} = -\frac{d}{ds} F(s)$ differentiation (13pt)

$\mathcal{L}\{e^{-t} f(t)\} = F(s+1)$ freq. shift (1pt)

$\mathcal{L}\{\cos(2t)\} = \frac{s}{s^2+4}$ cos: (1pt)

Let $f(s) = e^{-t} \cos(2t)$

$F(s) = \mathcal{L}\{e^{-t} \cos(2t)\} = \frac{s+1}{(s+1)^2+4}$ (13pt)

$z(s) = \mathcal{L}\{t e^{-t} \cos(2t)\} = -\frac{d}{ds} \left(\frac{s+1}{(s+1)^2+4} \right)$

$z(s) = \frac{-2(s+1)(s+1)}{[(s+1)^2+4]^2} = \frac{(s+1)^2-4}{[(s+1)^2+4]^2}$ (1pt)

$z(s) = \frac{(s-1)(s+3)}{[(s+1)^2+4]^2}$

Zeros

$s = 1, -3$ (1pt)

Poles

$s = -1 \pm 2j$ with multiplicity of 2

(1pt) each (1pt) 13pt → 7

+ 1pt trying

2. (20 points) Inverse Laplace Transform

Find the inverse Laplace transforms for the following signals.

a. (10 points) $X(s) = \frac{20e^{-2s}}{s(s^2+9s+20)}$

Let $X_1(s) = \frac{20}{s(s^2+9s+20)} = \frac{20}{s(s+4)(s+5)}$ — (A)

Computing partial fractions, $X_1(s) = \frac{1}{s} - \frac{5}{(s+4)} + \frac{4}{(s+5)}$ — (B)

The inverse Laplace transform of each of the terms is

$\frac{1}{s} \xleftrightarrow{\mathcal{L}} u(t)$, $\frac{5}{(s+4)} \xleftrightarrow{\mathcal{L}} 5e^{-4t}u(t)$ and $\frac{4}{(s+5)} \xleftrightarrow{\mathcal{L}} 4e^{-5t}u(t)$
 (C) (D) (E)

Combining, $X_1(s) \xleftrightarrow{\mathcal{L}} x_1(t) = u(t) - 5e^{-4t}u(t) + 4e^{-5t}u(t)$ — (F)

Using the time shifting property to include the effect of e^{-2s} ,

(G) $x(t) = u(t-2) - 5e^{-4(t-2)}u(t-2) + 4e^{-5(t-2)}u(t-2)$ — (H)

2 points for trying. 1 point for A-H.

b. (10 points) $X(s) = \frac{s^2+9s+17}{s^2+8s+16}$ — (A)

The order of the numerator is equal to the order of the denominator. Therefore, perform long division.

$X(s) = \frac{s^2+9s+17}{s^2+8s+16} = \frac{s^2+8s+16+s+1}{s^2+8s+16} = 1 + \frac{s+1}{s^2+8s+16}$ — (B)

Factoring the denominator gives

$X(s) = 1 + \frac{s+1}{(s+4)^2}$ (C)

Separating into Partial fractions:

$X(s) = 1 + \frac{s+4-3}{(s+4)^2} = 1 + \frac{s+4}{(s+4)^2} - \frac{3}{(s+4)^2} = 1 + \frac{1}{(s+4)} - \frac{3}{(s+4)^2}$
 (D) (E) (F)

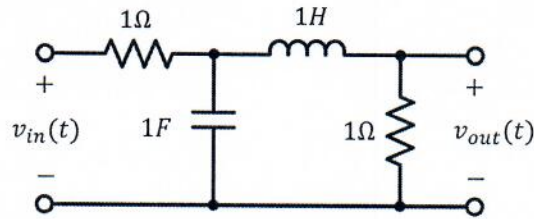
Finding the inverse of each term gives

$x(t) = \delta(t) + e^{-4t}u(t) - 3te^{-4t}u(t)$
 (G) (H)

2 points for trying. 1 point for A-H.

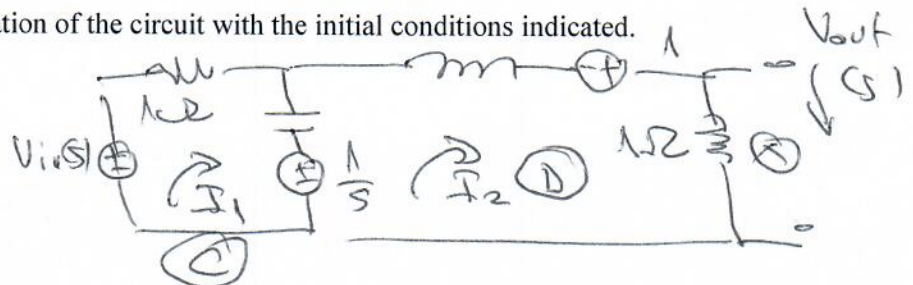
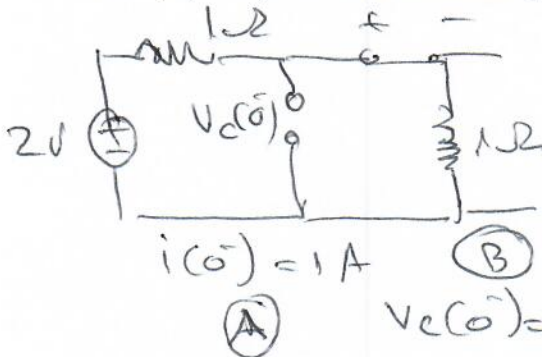
3. (20 points) Circuit Analysis

For the circuit below, the input $v_{in}(t) = x(t)$ is given in the graph. The units for $x(t)$ is Volts [V]. $v_{in}(t) = 2V$ for $t < 0$, and $v_{in}(t) = 2(1 - e^{-t+4})u(t - 4)$ for $t > 0$.



2 B, C D (p+)
1 p+ for trying

- a. (5 points) Draw the s-domain representation of the circuit with the initial conditions indicated.



1 p+ for trying

- b. (5 points) Find the s-domain representation of the input $v_{in}(t)$ for $t > 0$ given above.

$$V_{in}(s) = \mathcal{L}\{2(1 - e^{-t+4})u(t-4)\}$$

$$= 2 \cdot \left(\frac{1}{s} - \frac{1}{s+1} \right) e^{-4s} = \frac{e^{-4s} \cdot 2}{s(s+1)}$$

1 p+ 1 p+ 1 p+ 1 p+ 4+1=5

- c. (5 points) Write the mesh current equations in the s-domain needed to find the output $V_{out}(s)$. Express your answer by filling in the entries in the Cramer's method matrices shown below. Here, I_1 and I_2 represent the mesh currents (left to right), $V_1 = V_{in}(s) - \frac{1}{s}$ and $V_2 = V_{out}(s)$ represent the mesh voltages.

$$\begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

$$I_1 \left(1 + \frac{1}{s} \right) + I_2 \left(-\frac{1}{s} \right) = V_{in}(s) - \frac{1}{s}$$

$$I_1 \left(-\frac{1}{s} \right) + I_2 \left(\frac{1}{s} + s + 1 \right) = \frac{1}{s} + 1$$

A₃ A₄ V₂

1 p+ for trying

A_{1,2,3,4} (2 p+)

V₁ (1 p+)
V₂ (1 p+)

$$\begin{pmatrix} 1 - \frac{1}{s} & -\frac{1}{s} \\ -\frac{1}{s} & \frac{1}{s} + s + 2 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} V_{in}(s) - \frac{1}{s} \\ \frac{1}{s} + 1 \end{pmatrix}$$

1 pt
for
full
matrix

$$\begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

d. (5 points) Write the formula for $V_{out}(s)$ as a ratio of polynomials.

$$\Delta = A_1 A_4 - A_2 A_3 = s + \frac{2}{s} + 2$$

$$\Delta I_2 = \left(1 + \frac{1}{s}\right) \left(\frac{1}{s} + 1\right) + \frac{1}{s} \left(V_{in} - \frac{1}{s}\right) = \frac{V_{in} + s + 2}{s}$$

$$V_{out} = I_2(s) \cdot 1\Omega \quad \otimes \quad 1\Omega \text{ resistor} \quad \boxed{1 \text{ pt}}$$

$$\boxed{1 \text{ pt}} \quad V_{out} = \frac{V_{in}(s) + s + 2}{s^2 + 2s + 2} = \frac{\frac{2e^{-4s}}{s(s+1)} + s + 2}{s^2 + 2s + 2} =$$

$$= \frac{2e^{-4s} + (s+2)(s+1) + s}{s(s+1)(s^2 + 2s + 2)} \quad \boxed{1 \text{ pt}} \quad \boxed{1 \text{ pt}} \text{ for trying}$$

4. (20 points) System Response Partitions

A linear time-invariant system is described by the following LCCDE

$$\frac{d^2 y(t)}{dt^2} + 6 \frac{dy(t)}{dt} + 9y(t) = x(t)$$

The input is $x(t) = 2e^{-t}\cos(t)u(t)$, with the initial conditions $y(0^-) = 2$, $y'(0^-) = -12$.

- a. (5 points) Derive a formula for the output $Y(s)$. Your answer should be expressed as a ratio of polynomials with a fully factored denominator.

$$\begin{aligned} y(t) &\stackrel{\mathcal{L}}{\leftrightarrow} Y(s) \\ \frac{d}{dt}y(t) &\stackrel{\mathcal{L}}{\leftrightarrow} sY(s) - y(0^-) \\ \frac{d^2}{dt^2}y(t) &\stackrel{\mathcal{L}}{\leftrightarrow} s^2Y(s) - sy(0^-) - y'(0^-) \end{aligned}$$

Therefore, taking the Laplace transform of the LCCDE yields:

$$\begin{aligned} s^2Y(s) - sy(0^-) - y'(0^-) + 6(sY(s) - y(0^-)) + 9Y(s) &= X(s) \\ s^2Y(s) + 6sY(s) + 9Y(s) - sy(0^-) - y'(0^-) - 6y(0^-) &= X(s) \end{aligned}$$

— (A)

Using the given initial conditions

$$\begin{aligned} s^2Y(s) + 6sY(s) + 9Y(s) - 2s + 12 - 12 &= X(s) \\ s^2Y(s) + 6sY(s) + 9Y(s) &= X(s) + 2s \end{aligned}$$

$$Y(s)(s^2 + 6s + 9) = X(s) + 2s$$

— (B)

Also,

$$\begin{aligned} x(t) &\stackrel{\mathcal{L}}{\leftrightarrow} X(s) \\ 2e^{-t}\cos(t)u(t) &\stackrel{\mathcal{L}}{\leftrightarrow} \frac{2(s+1)}{(s+1)^2 + 1} \end{aligned}$$

Substituting $X(s) = \frac{2(s+1)}{(s+1)^2 + 1}$, (C)

$$Y(s)(s^2 + 6s + 9) = \frac{2(s+1)}{(s+1)^2 + 1} + 2s = \frac{2(s+1) + 2s[(s+1)^2 + 1]}{(s+1)^2 + 1}$$

$$Y(s) = \frac{2(s+1) + 2s[(s+1)^2 + 1]}{((s+1)^2 + 1)(s^2 + 6s + 9)}$$

$$Y(s) = \frac{2(s+1) + 2s[(s+1)^2 + 1]}{((s+1)^2 + 1)(s+3)^2}$$

— (D)

Alternate form: $Y(s) = \frac{2(s+1) + 2s[(s+1)^2 + 1]}{(s+1+j)(s+1-j)(s+3)^2}$

1 point for trying. 1 point for A-D.

- b. (5 points) Calculate the final value $y(\infty)$ of this system's response.

To find the final value, we can use the final value theorem. (A)

$$y(\infty) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} s \frac{2(s+1) + 2s[(s+1)^2 + 1]}{((s+1)^2 + 1)(s+3)^2} \quad \text{--- (C)}$$

The limit is 0 since the numerator goes to 0, but the denominator does not go to 0. (D)

1 point for trying. 1 point for A-D.

- c. (5 points) Calculate the zero-input response of the above system in the **time** domain.

$$Y(s)(s^2 + 6s + 9) = X(s) + 2s$$

For the zero-input response, substitute $X(s) = 0$ (A)

$$\begin{aligned} Y(s)(s^2 + 6s + 9) &= 2s \\ Y(s) &= \frac{2s}{(s^2 + 6s + 9)} \quad \text{--- (B)} \\ &= \frac{2s}{(s+3)^2} \\ &= \frac{2(s+3-3)}{(s+3)^2} = \frac{2}{(s+3)^2} - \frac{6}{(s+3)^2} \quad \text{--- (C)} \end{aligned}$$

Taking the inverse Laplace transform of each term:

$$y(t) = 2e^{-3t}u(t) - 6te^{-3t}u(t) \quad \text{--- (D)}$$

1 point for trying. 1 point for A-D.

- d. (5 points) Calculate the zero-state response of the above system in the **Laplace** domain. (A)

For the zero-state response, make the initial conditions zero, i.e. $y(0^-) = 0$, $y'(0^-) = 0$. Therefore,

$$\begin{aligned} Y(s)(s^2 + 6s + 9) &= X(s) \\ Y(s) &= \frac{X(s)}{(s^2 + 6s + 9)} \quad \text{--- (B)} \\ x(t) = 2e^{-t}\cos(t)u(t) &\stackrel{\mathcal{L}}{\leftrightarrow} \frac{2(s+1)}{(s+1)^2 + 1} \\ X(s) &= \frac{2(s+1)}{(s+1)^2 + 1} \quad \text{--- (C)} \end{aligned}$$

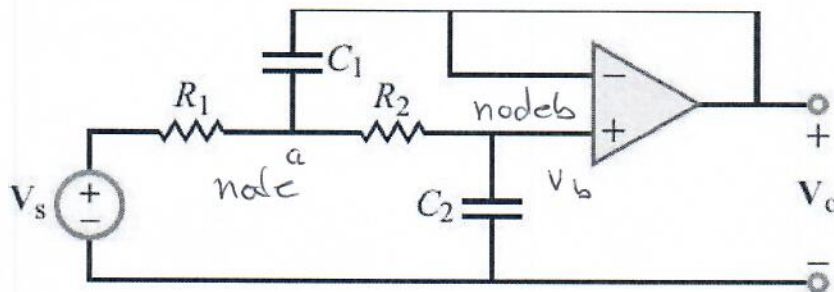
Therefore,

$$Y(s) = \frac{2(s+1)}{[(s+1)^2 + 1](s^2 + 6s + 9)} \quad \text{--- (D)}$$

1 point for trying. 1 point for A-D.

5. (10 points) Operational Amplifiers

(1 pt) trying



(1 pt)
 $V_P = V_N$
 negative feedback

a. (6 points) Calculate the transfer function $H(s)$ for this circuit.

(1 pt) KCL node a

$$\frac{V_s - V_a}{R_1} = \frac{V_o - V_b}{R_2} + \frac{(V_a - V_b)}{\frac{1}{sC_1}}$$

(1 pt) KCL node b

$$\frac{V_b}{\frac{1}{sC_2}} = \frac{V_a - V_b}{\frac{1}{R_2} + sC_1}$$

eliminate V_a

$$V_a = V_b \left(1 + \frac{sC_2 R_2}{1 + sR_2 C_1} \right) \quad H(s) = \frac{1 + sR_2 C_1}{1 + s(R_1 C_2 + R_2 C_2 + R_2 C_1) + s^2 R_1 R_2 C_1 C_2}$$

~~(1 pt)~~

(1 pt)

(1 pt)

□ 1 pt + 1 pt trying

b. (4 points) Identify the poles and zeros of the transfer function $H(s)$ for $R_1 = R_2 = 1\Omega$ and $C_1 = C_2 = 1F$.

$$H(s) = \frac{1 + s \cdot 1 \cdot 1}{s^2 + 1 \cdot 1 \cdot 1 \cdot 1 + s[1(1+1) + 1 \cdot 1] + 1} = \frac{1 + s}{s^2 + s(2+1) + 1}$$

VPT
 100 trying

(1 pt)

$$= \frac{s+1}{s^2+3s+1}$$

2 roots $s = -1$ (1 pt)

(2)

poles

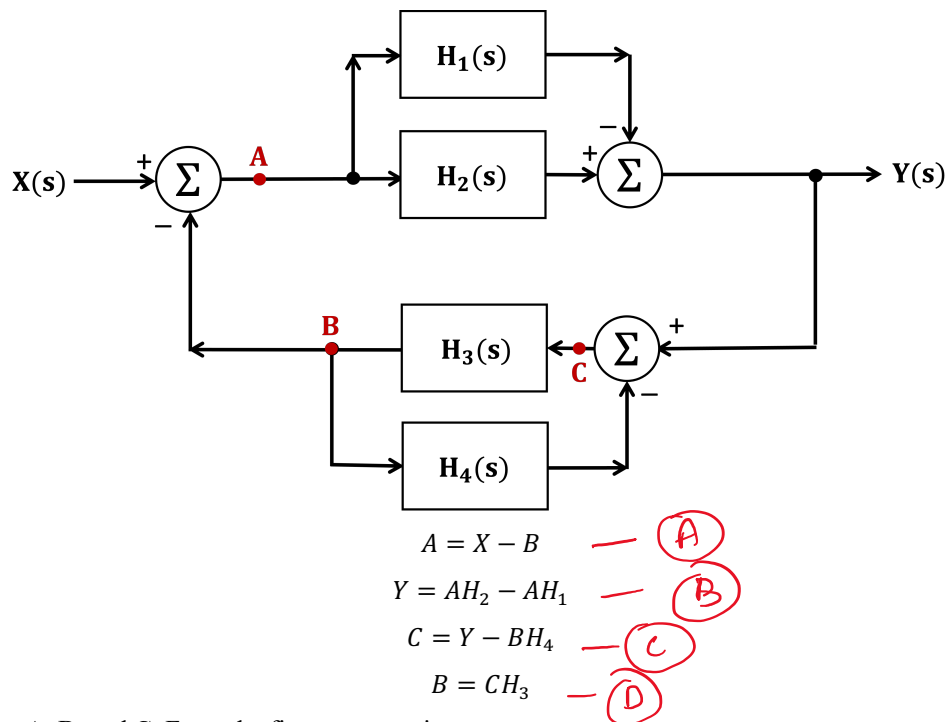
$$s = \frac{-3 \pm \sqrt{3^2 - 4 \cdot 1 \cdot 1}}{2} = \frac{-3 \pm \sqrt{5}}{2}$$

(2)

(2+2) → 4

6. (10 points) System Block Diagrams

Calculate the overall transfer function $\mathbf{H(s)}$ for the following connection of LTI systems. Your final answer should be expressed in the Laplace domain.



Substitute and eliminate A, B, and C. From the first two equations,

$$Y = A(H_2 - H_1) = (X - B)(H_2 - H_1)$$

From the last two equations,

$$B = (Y - BH_4)H_3$$

$$B + BH_4H_3 = YH_3$$

$$B(1 + H_4H_3) = YH_3$$

$$B = \frac{YH_3}{(1 + H_4H_3)}$$

Combining

$$Y = \left(X - \frac{YH_3}{(1 + H_4H_3)} \right) (H_2 - H_1)$$

$$Y + Y \frac{H_3(H_2 - H_1)}{(1 + H_4H_3)} = X(H_2 - H_1)$$

$$Y = X \frac{(H_2 - H_1)}{\left[1 + \frac{H_3(H_2 - H_1)}{(1 + H_4H_3)} \right]}$$

The transfer function is $H = \frac{Y}{X}$

$$H = \frac{(H_2 - H_1)}{\left[1 + \frac{H_3(H_2 - H_1)}{(1 + H_4H_3)} \right]}$$

2 points for trying. 1 point for A-F. 2 points for G.